En:-25084116002

PRACTICAL:-3

AIM:- Apply ltering, sorting, aggregation, and GroupBy operations on real-world datasets.

Step 1: Display Dataset

```
# Import Library.
import pandas as pd
# Create Dataset.
df = {
       "EMP_ID": [101, 102, 103, 104, 105, 106, 107, 108],
       "Name": ["Rahul", "Priya", "Amit", "Riya", "Karan", "Neha", "Vivek", "Pooja"],
       "Department": ["IT", "HR", "IT", "Finance", "HR", "Finance", "IT", "HR"],
       "City": ["Surat", "Navasari", "Surat", "Valsad", "Surat", "Navasari", "Valsad", "Surat"],
       "Salary": [45000, 38000, 52000, 48000, 40000, 51000, 60000, 42000],
       "Experience": [2, 1, 5, 4, 2, 6, 8, 3]
       }
df=pd.DataFrame(df)
df
```

	EMP_ID	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
1	102	Priya	HR	Navasari	38000	1
2	103	Amit	IT	Surat	52000	5
3	104	Riya	Finance	Valsad	48000	4
4	105	Karan	HR	Surat	40000	2
5	106	Neha	Finance	Navasari	51000	6
6	107	Vivek	IT	Valsad	60000	8
7	108	Pooja	HR	Surat	42000	3

Next steps: Generate code with df | New interactive sheet

```
# Display employees from IT department.
df[df['Department'] == 'IT']
```

	EMP_ID	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

```
# Display employees with salary > 50000.
df[df['Salary'] > 50000]
```

	EMP_ID	Name	Department	City	Salary	Experience
2	103	Amit	IT	Surat	52000	5
5	106	Neha	Finance	Navasari	51000	6
6	107	Vivek	IT	Valsad	60000	8

```
# Display employees from Surat.
df[df['City'] == 'Surat']
```

	EMP_ID	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
2	103	Amit	IT	Surat	52000	5
4	105	Karan	HR	Surat	40000	2
7	108	Pooja	HR	Surat	42000	3

```
# Display employees from IT department having salary > 5000
df[(df['Department'] == 'IT') & (df['Salary'] > 50000)]
```

	EMP_ID	Name	Department	City	Salary	Experience
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

```
# Salary ascending
df.sort_values('Salary', ascending=True)
```

	EMP_ID	Name	Department	City	Salary	Experience
1	102	Priya	HR	Navasari	38000	1
4	105	Karan	HR	Surat	40000	2
7	108	Pooja	HR	Surat	42000	3
0	101	Rahul	IT	Surat	45000	2
3	104	Riya	Finance	Valsad	48000	4
5	106	Neha	Finance	Navasari	51000	6
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

```
# Salary descending
df.sort_values('Salary', ascending=False)
```

	EMP_ID	Name	Department	City	Salary	Experience
6	107	Vivek	IT	Valsad	60000	8
2	103	Amit	IT	Surat	52000	5
5	106	Neha	Finance	Navasari	51000	6
3	104	Riya	Finance	Valsad	48000	4
0	101	Rahul	IT	Surat	45000	2
7	108	Pooja	HR	Surat	42000	3
4	105	Karan	HR	Surat	40000	2
1	102	Priya	HR	Navasari	38000	1

```
#  department wise salary in descending
df.sort_values(by=['Department', 'Salary'],ascending=[True, False])
```

	EMP_ID	Name	Department	City	Salary	Experience
5	106	Neha	Finance	Navasari	51000	6
3	104	Riya	Finance	Valsad	48000	4
7	108	Pooja	HR	Surat	42000	3
4	105	Karan	HR	Surat	40000	2
1	102	Priya	HR	Navasari	38000	1
6	107	Vivek	IT	Valsad	60000	8
2	103	Amit	IT	Surat	52000	5
0	101	Rahul	IT	Surat	45000	2

```
# Calculate the total salary.
df['Salary'].sum()
```

```
np.int64(376000)
```

```
# Calculate thye avg salary of employees.
df['Salary'].mean()
```

```
np.float64(47000.0)
```

```
# Display max salary.
df['Salary'].max()
```

```
60000
```

```
# Display lowest salary.
df['Salary'].min()
```

```
38000
```

```
#Count the total no of employees.
df['EMP_ID'].count()
```

```
np.int64(8)
```

```
# Average salary of each department.
df.groupby('Department')['Salary'].mean().reset_index()
```

	Department	Salary
0	Finance	49500.000000
1	HR	40000.000000
2	IT	52333.333333

```
# Total salary of each department.
df.groupby('Department')['Salary'].sum().reset_index()
```

	Department	Salary
0	Finance	99000
1	HR	120000
2	IT	157000

```
# Display the higest salary in each department.
df.groupby('Department')['Salary'].max().reset_index()
```

	Department	Salary
0	Finance	51000
1	HR	42000
2	IT	60000

```
# Count the no of employees in dept.
df.groupby('Department')['EMP_ID'].count().reset_index()
```

	Department	EMP_ID
0	Finance	2
1	HR	3
2	IT	3

```
# Calculate the avg salary of employee.
df.groupby('City')['Salary'].mean().reset_index()
```

	City	Salary
0	Navasari	44500.0
1	Surat	44750.0
2	Valsad	54000.0

```
# Apply multiple statistical function.
df.groupby('Department')['Salary'].agg(['sum', 'mean', 'max', 'min', 'count']).reset_index()
```

Department	sum	mean	max	min	count

```
# Save CSV file.
df.to_csv("Employee_Data.csv", index=False)
print("CSV file saved Successfully")
```

```
CSV file saved Successfully
```

```
# Import the files module from Google Colab.
from google.colab import files
files.download("Employee_Data.csv")
```